- **Explain what the transformer does to the HTML retrieved by the loader**

It walks through the html and extracts every text that is contained within an article, and “cleans” it, which is then returned to the calling function

- **Examine the document cleaning code. What kinds of characters are removed when the text is cleaned?**

The clean_text function mainly just removes leading and trailing white space, multiple line spacing, and accents/special characters.

- **What size chunks is the content split into?**

10000 characters

- **What amount of overlap is there between chunks?**

1000 characters

- **How many documents are loaded at a time into the vector database?**

300

- **Tell me about the MS program**

  https://www.pdx.edu/computer-science/master

- **Tell me about the Cybersecurity certificate**

  https://www.pdx.edu/computer-science/cybersecurity

- **Tell me about PCEP**

  https://www.pdx.edu/computer-science/pcep-internship

- **Tell me about the capstone**

  https://www.pdx.edu/computer-science/bachelors-program-computer-science-capstone

- **How many credits does it take to complete the MS program?**

  45 credits

```
llm>> How many credits does it take to complete the MS program?
It takes 45 credits to complete the MS program.  There is also a thesis option.

For more information, please visit these URLs:

* https://www.pdx.edu/computer-science/cybersecurity
* https://www.pdx.edu/computer-science/master
* https://www.pdx.edu/computer-science/honors-track
llm>>
```

- **How many credits does it take to complete the MS program?**

```
llm>> How many core courses are there in the Cybersecurity certificate?
There are two core courses in the Cybersecurity Graduate Certificate program.  These courses total 6 credit hours.

For more information, please visit the following URLs:

* [https://www.pdx.edu/computer-science/graduate](https://www.pdx.edu/computer-science/graduate)
* [https://www.pdx.edu/computer-science/masters-track-courses](https://www.pdx.edu/computer-science/masters-track-cours
s)
* [https://www.pdx.edu/computer-science/cybersecurity](https://www.pdx.edu/computer-science/cybersecurity)
llm>>
```

- **How many hours per week does a student work in PCEP?**

```
* [https://www.pdx.edu/computer-science/cybersecurity](https://www.pdx.edu/c
llm>> How many hours per week does a student work in PCEP?
Students in the PCEP program work 20 hours per week.

For more information, please visit these URLs:

* https://www.pdx.edu/computer-science/pcep-internship
* https://www.pdx.edu/computer-science/internships-and-research
llm>>
```

- **How long is the capstone?**

```
* https://www.pdx.edu/computer-science/internships-and-research
llm>> How long is the capstone?
The capstone project lasts for two quarters, which is six months.

For more information, please visit the following URLs:

* Capstone home page: https://sites.google.com/pdx.edu/bruceirvin/capstone
* Guidelines for Sponsors: https://sites.google.com/pdx.edu/bruceirvin/capstone
* Bachelor's Program in Computer Science Capstone: https://www.pdx.edu/computer-science/bachelors-program-co
ce-capstone
* Undergraduate Courses: https://www.pdx.edu/computer-science/undergraduate-courses
llm>> |
```

- **What is the current xkcd?**

```
> Finished chain.

Observation: The current XKCD comic is number 3061, titled "Water Balloons". The image URL is https:
cs/water_balloons.png, the alt text is "Update: The physics department has recruited an astronomer w
reballs.", and there is no transcript.
Thought:I am finished executing a plan and have the information the user asked for.
Final Answer: The current XKCD comic is number 3061, titled "Water Balloons".  The image URL is http
mics/water_balloons.png, the alt text is "Update: The physics department has recruited an astronomer
fireballs.", and there is no transcript.

> Finished chain.
{'input': 'What is the current xkcd?', 'output': 'The current XKCD comic is number 3061, titled "Wat
mage URL is https://imgs.xkcd.com/comics/water_balloons.png, the alt text is "Update: The physics de
ed an astronomer who studies meteor fireballs.", and there is no transcript.'}
llm>> |
```

- **What is the image link of the current xkcd?**

https://imgs.xkcd.com/comics/water_balloons.png,

- **What was xkcd 327 about?**

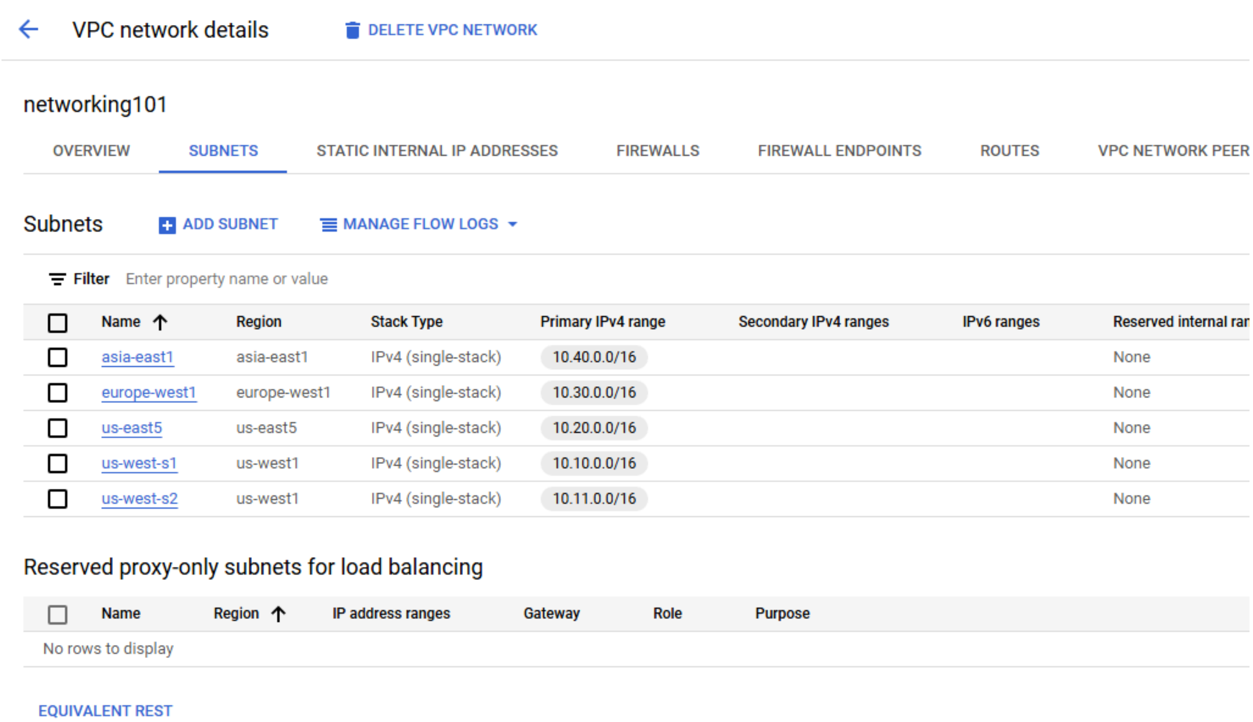
```
> Finished chain.
{'input': 'What was xkcd 327 about?', 'output': 'xkcd 327 was titled "Exploits of a Mom".  The alt te
r is named Help I\'m trapped in a driver\'s license factory."  The image URL is https://imgs.xkcd.com
_a_mom.png'}
llm>> |
```

- **Take a screenshot of the output to include in your lab notebook. How many networks, subnetworks, and VM instances have been created?**

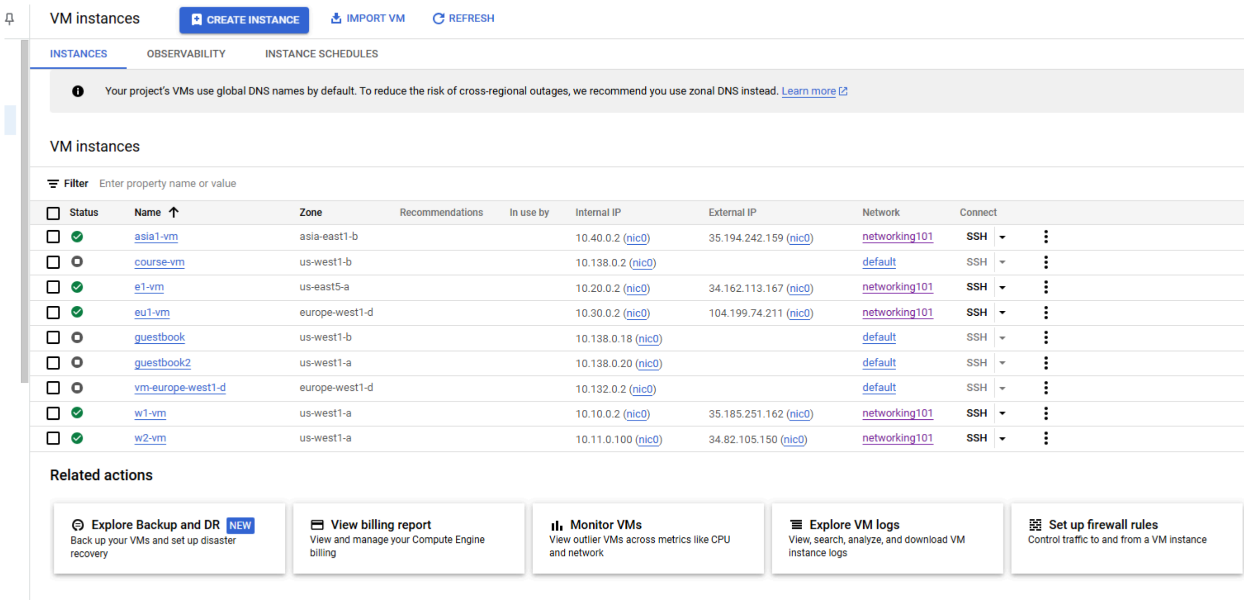
1 network, 5 subnetworks, and 5 VM instances

```
NAME: asia-east1
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: asia1-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: e1-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: eu1-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: europe-west1
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: networking101
TYPE: compute.v1.network
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: us-east5
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: us-west-s1
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: us-west-s2
TYPE: compute.v1.subnetwork
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: w1-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:

NAME: w2-vm
TYPE: compute.v1.instance
STATE: COMPLETED
ERRORS: []
INTENT:
```

- **Visit the web console for VPC network and show the network and the subnetworks that have been created. Validate that it has created the infrastructure in the initial figure. Note the lack of firewall rules that have been created.**



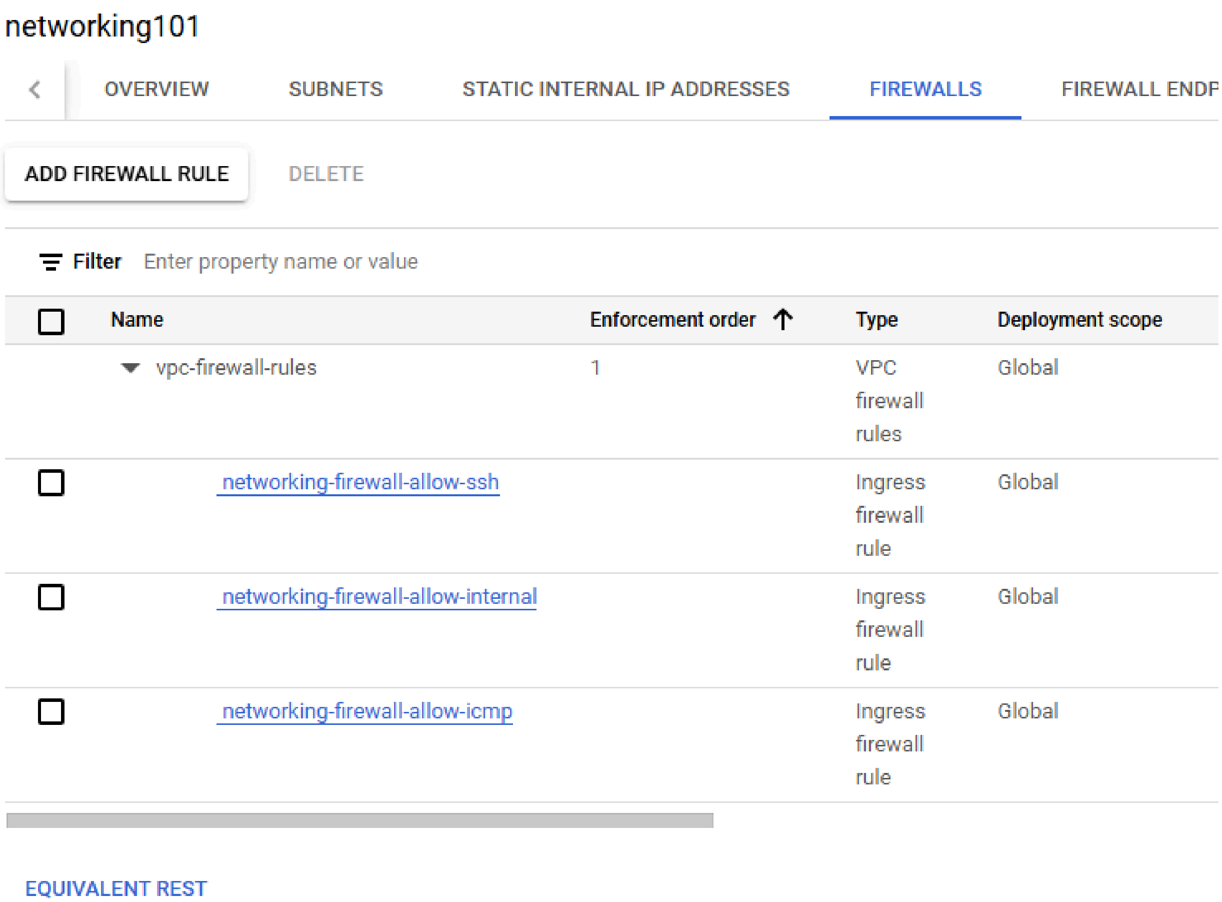
- **Visit the web console for Compute Engine and show all VMs that have been created, their internal IP addresses and the subnetworks they have been instantiated on. Validate that it has created the infrastructure shown in the initial figure.**



- **Click on the ssh button for one of the VMs and attempt to connect. Did it succeed?**

No! It is stuck on “Establishing connection to SSH server…”

- **Take a screenshot that indicates the new rules have been deployed**



- **Given this, fill in the table with the measured latencies between the 6 pairs and include it in your lab notebook. Use the shortest latency measured for each pair.**

Location pair	Ideal latency	Measured latency
us-west1 us-east5	~45 ms	53.9ms
us-west1 europe-west1	~93 ms	102ms
us-west1 asia-east1	~114 ms	116 ms
us-east5 europe-west1	~76 ms	91.4ms

us-east5 asia-east1	~141 ms	174 ms
europe-west1 asia-east1	~110 ms	264 ms

- **Are the instances in the same availability zone or in different ones?**

  Different!

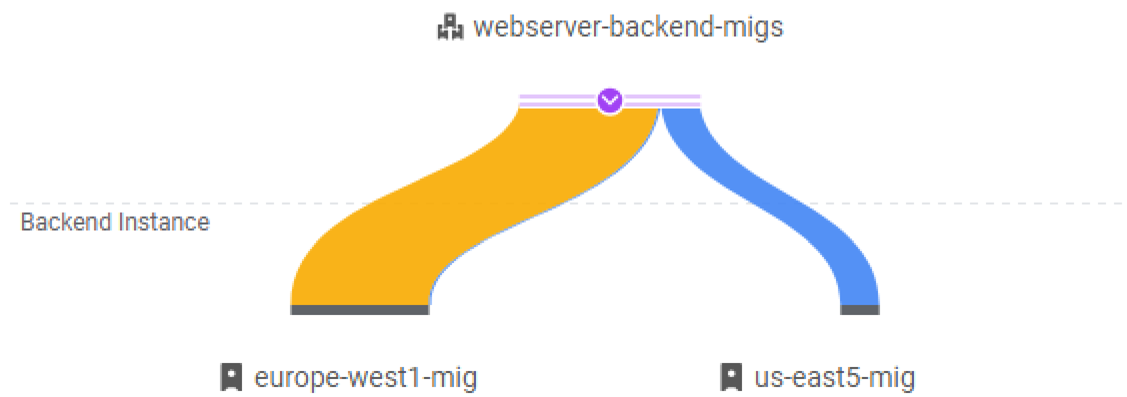
- **List all availability zones that your servers show up in for your lab notebook.**

Europe-west1-c, europe-west1-b, europe-west1-d, us-east-5b

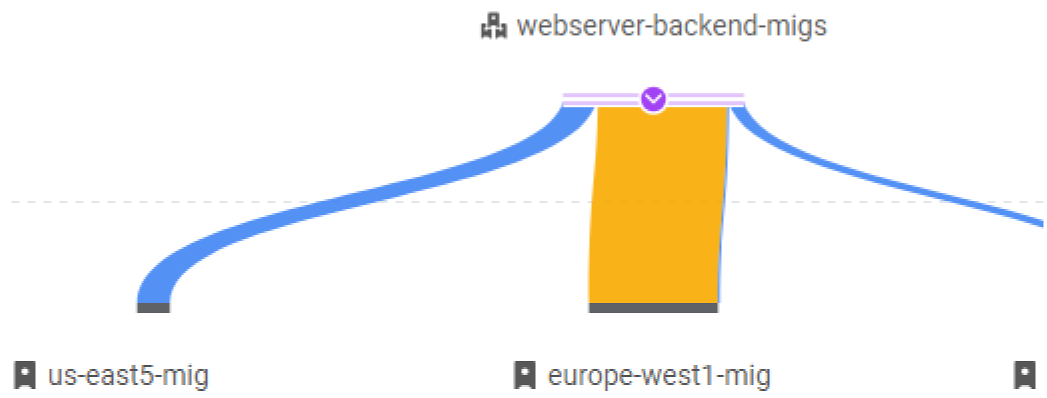
- **Which availability zone does the server handling your request reside in?**

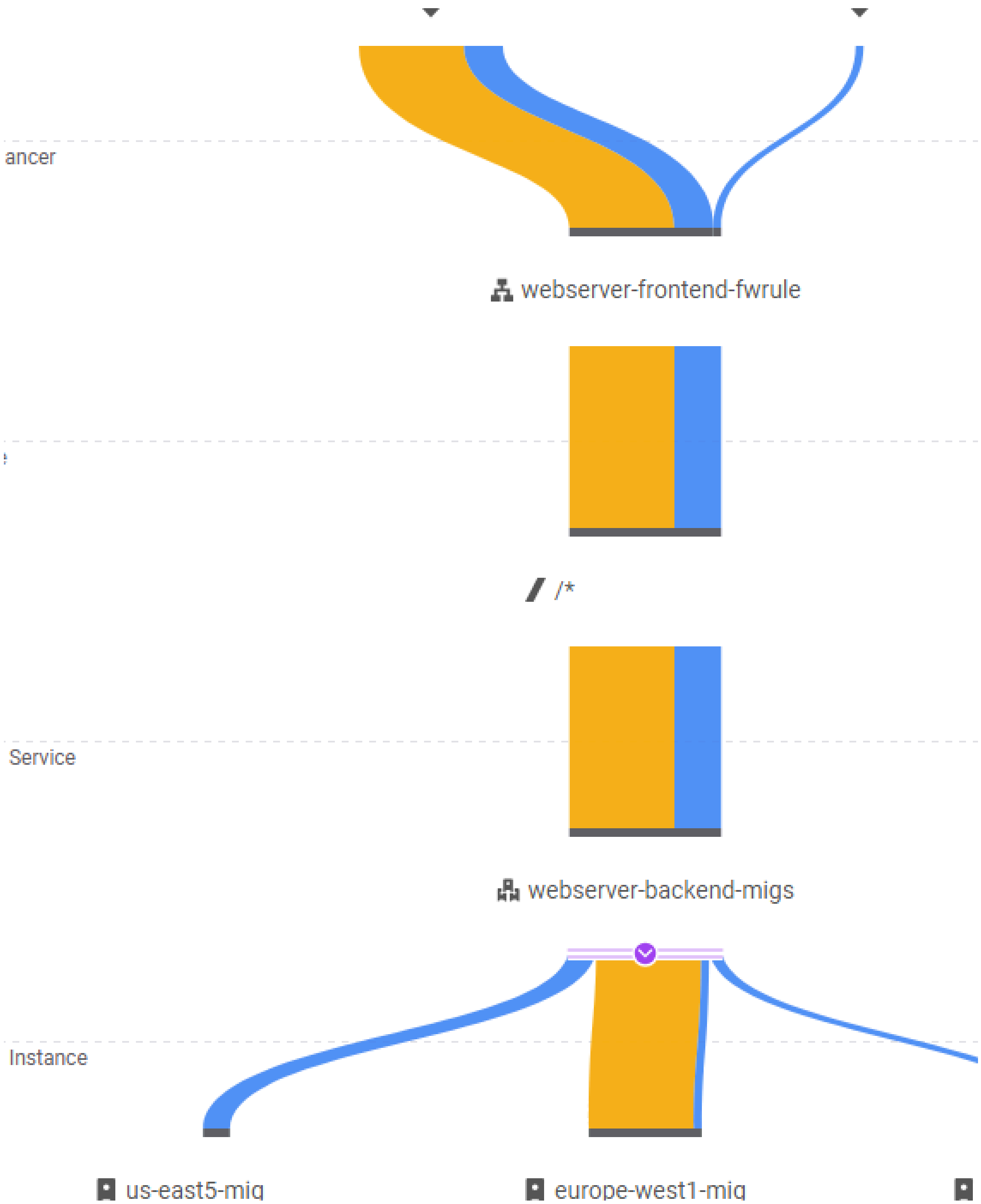
Us-east-5b

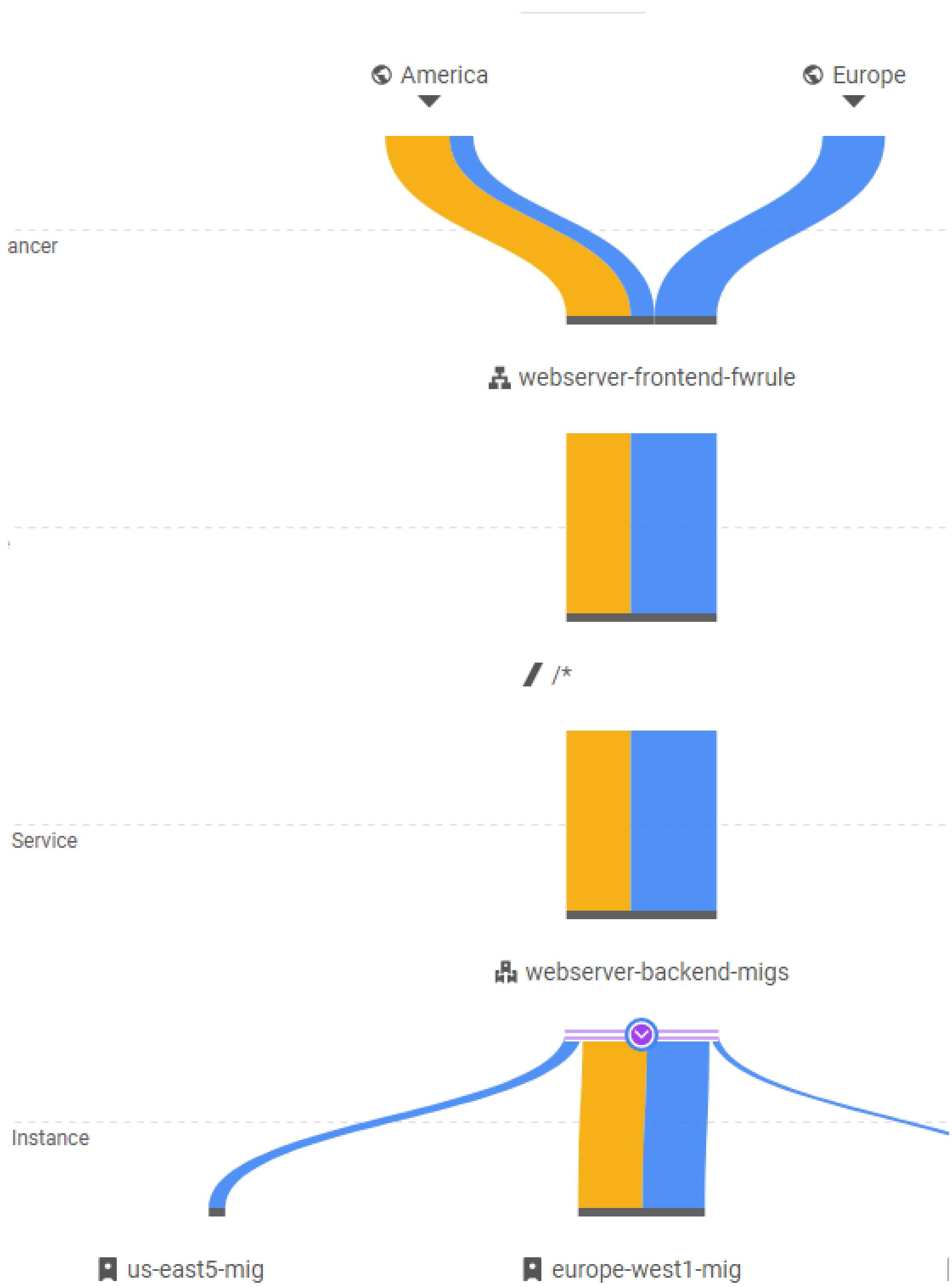
- **Take a screenshot of the initial traffic distribution**



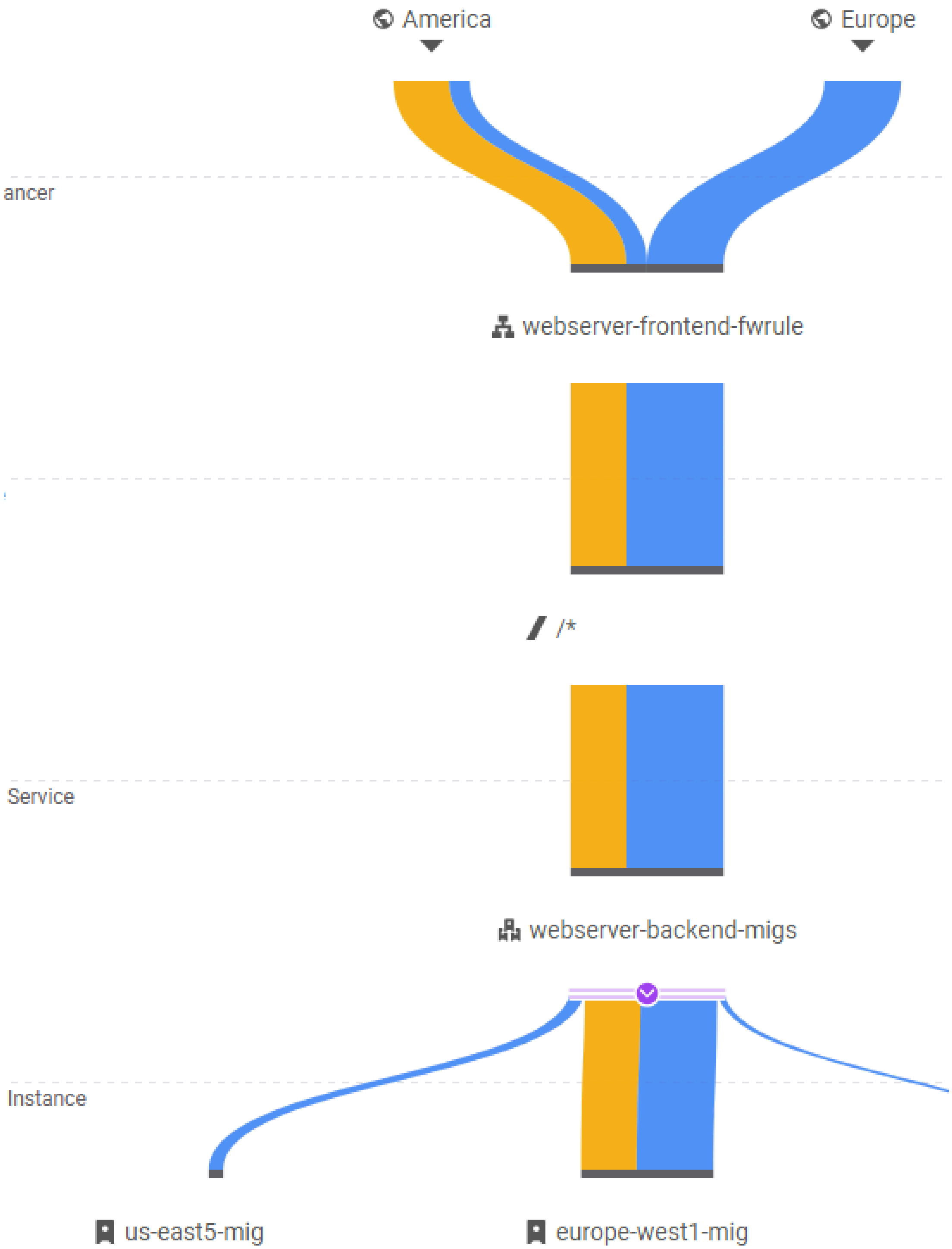
- **Take a screenshot of the UI as additional instances are brought up and show that the traffic distribution shifts**


webserver-backend-migs
us-east5-mig
europe-west1-mig


ancer
webserver-frontend-fwrule
/*
Service
webserver-backend-migs
Instance
us-east5-mig
europe-west1-mig



America
Europe
ancer
webserver-frontend-fwrule
/*
Service
webserver-backend-migs
Instance
us-east5-mig
europe-west1-mig


America
Europe
ancer
webserver-frontend-fwrule
/*
Service
webserver-backend-migs
Instance
us-east5-mig
europe-west1-mig